Lab 3

Relational & Logical operators and conditional

Learning Objectives

The following types will be covered in this lab session

- To understand relational operators in C++.
- To learn logical operators and their use in decision making.
- To implement conditional statements (if, if-else, nested if).
- To solve problems using conditions in programming.

Relational Operators

Relational operators are used to compare two values. The result of a relational operation is either true (1) or false (0). These operators are mainly used in decision-making and loops.

They help the program decide conditions like whether one value is greater, smaller, or equal to another.

Syntax:

```
a == b   // equal to
a != b   // not equal to
a > b    // greater than
a < b    // less than
a >= b   // greater than or equal to
a <= b   // less than or equal to
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5;   // Declare variables
    cout << (a > b) << endl;   // 1 (true)
    cout << (a < b) << endl;   // 0 (false)
    cout << (a == b) << endl;  // 0 (false)
    return 0; }
```

2. Logical Operators

logical operators are used to combine multiple conditions. They are used when we need to check more than one condition at a time in decision-making.

They return either true or false based on combined conditions.

Syntax:

```
&&   // AND (both conditions must be true)

||      // OR (any one condition true)

!      // NOT (reverses condition)
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5;
    cout << (a > 5 && b < 10) << endl;  // 1 (true)
    cout << (a < 5 || b < 10) << endl;  // 1 (true)
    cout << !(a > b) << endl;           // 0 (false)
    return 0;
}
```

3. Conditional Statements (if, if-else, nested if)

if Statement:

The if statement is used to execute a block of code only when a condition is true. If the condition is false, the block is skipped

Syntax:

```
if (condition) {
    // code executes if condition is true
}
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int num = 10;
    if (num > 0) {
        cout << "Number is positive" << endl;
    }
    return 0;
}
```

If-else- Statement:

The if-else statement is used when we have two choices. If the condition is true, one block executes; otherwise, the else block executes.

Syntax:

```
if (condition) {
    // true block
} else {
    // false block
}
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    if (num % 2 == 0) {
        cout << "Even number" << endl;
    } else {
        cout << "Odd number" << endl;
    }
    return 0;
}
```

Nested If Statement:

A nested if is an if statement inside another if statement. It is used when multiple conditions depend on each other.

Syntax:

```
if (condition1) {

    if (condition2) {
        // code
    }
}
```

Example:

```
#include <iostream>
using namespace std;

int main() {

    int age = 20;

    if (age >= 18) {        //check if the condition true

        if (age <= 60) {   //check if the 2nd condition true
            cout << "Adult" << endl;
        }
    }   return 0;}
```

In this lab, we studied relational operators, logical operators, and conditional statements in C++. We learned how relational operators compare values, how logical operators combine conditions, and how conditional statements help in decision-making. These concepts are essential for building logical and real-world programs.

Lab 3: Tasks

Task 1: Comparison / Relational Operators

- Write a C++ Program that prompts the uses to enter two numbers
- Use comparison operators to check if the first number is equal to or greater than or less than second number
- Display appropriate messages based on the comparison result

```
#include <iostream>                  // Header file for input/output
using namespace std;                  // Standard namespace
int main(){                          // Main function
   int num1, num2;                    // Variables for two numbers
   cout << "Enter first number: ";   // Ask user
   cin >> num1;                      // Input first number
   cout << "Enter second number: ";  // Ask user
   cin >> num2;                       // Input second number
   if(num1 == num2)                  // Compare equal
      cout << "First number is equal to second number" : // Output
   if(num1 > num2)                   // Compare greater
      cout << "First number is greater than second number" << endl;
   if(num1 < num2)                   // Compare less
      cout << "First number is less than second number" << endl;
   return 0;    }                    // End program
```

Task 2: Logical Operators

- Write a C program that prompts the user to enter an integer. Use logical operators to check if the entered number is greater than 0 and less than 100.
- Display a message indicating whether the number meets both conditions or not.

```
#include <iostream>                      // Header file

using namespace std;                     // Standard namespace

int main()                               // Main function

{

    int a, b;                            // Variables

    cout << "Enter first number: ";      // Ask user

    cin >> a;                            // Input

    cout << "Enter second number: ";     // Ask user

    cin >> b;                            // Input

    if(a > 0 && b > 0)                   // AND operator

        cout << "Both numbers are positive" << endl; // Output

    if(a > 0 || b > 0)                   // OR operator

        cout << "At least one number is positive" << endl;

    if(!(a == b))                        // NOT operator

        cout << "Numbers are not equal" << endl;

    return 0;                            // End
```

Task 3: If Statement

- Write a C program that prompts the user to enter an integer.
  Use an if statement to check if the entered number is positive.
- Display a message indicating whether the number is positive or not.
-

```
#include <iostream>                    // Header file

using namespace std;                   // Standard namespace

int main()                             // Main function

{

    int num;                           // Variable

    cout << "Enter a number: ";        // Ask user

    cin >> num;                         // Input

    if(num > 0)                        // Check condition

        cout << "Number is positive" << endl; // Output

    return 0;                          // End

}
```

Task 4: If-Else Statement

- Write a C program that prompts the user to enter their age. Use an if-else statement to check if the entered age is greater than or equal to 18.
- Display appropriate messages for both conditions.

```
#include <iostream>                    // Header file

using namespace std;                   // Standard namespace

int main()                             // Main function

    int num;                           // Variable

    cout << "Enter a number: ";        // Ask user


    cin >> num;                        // Input

    if(num % 2 == 0)                   // Check even

        cout << "Number is even" << endl; // Output

    else                               // Otherwise

        cout << "Number is odd" << endl;

    return 0;                          // End

}
```

Task 5: Else-If Statement

- Write a C program that prompts the user to enter an integer. Use an else-if statement to check if the entered number is positive, negative, or zero.
- Display appropriate messages based on the entered number.

```
#include <iostream>                   // Header file
using namespace std;                  // Standard namespace
int main(){                           // Main function
    int marks;                        // Variable
    cout << "Enter marks: ";          // Ask user
    cin >> marks;                     // Input

    if(marks >= 80)                   // Condition 1
        cout << "Grade A" << endl;    // Output
    else if(marks >= 60)              // Condition 2
```

```
        cout << "Grade B" << endl;     // Output
    else if(marks >= 40)               // Condition 3
        cout << "Grade C" << endl;     // Output
    else                               // Otherwise
        cout << "Fail" << endl;        // Output
    return 0;                          // End
}
```